Data Analysis in Python

Mini Project Report

A Model Submitted

in Partial Fulfilment of the Requirements for the Degree of

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in

CSE

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1. Objective

This mini project aims to perform end-to-end data analysis using Python on the CarDekho used car dataset. The primary goal was to understand price patterns, usage factors, and categorize cars using machine learning clustering (K-Means). The project includes data cleaning, EDA, statistical testing, clustering, and interpretation.

2. Data Set Description

The dataset contains detailed information about used cars, including:
- Selling price
- Manufacturing year
- Kilometres driven
- Fuel type
- Transmission type
- Owner type

This dataset is commonly sourced from Car Dekho and represents real-world automobile market conditions.

3. Methods Used

3.1 Data Preprocessing

- Removal of duplicate entries.
- Creation of new feature 'Age' from manufacturing year.
- Handling outliers using the IQR method.
- Label Encoding for fuel type, transmission, and owner.
- Standard scaling of numerical features for K-Means clustering.

3.2 Exploratory Data Analysis (EDA)

- Distribution of price, km driven, and age analyzed using histograms and boxplots
- Price vs Age and Price vs Km Driven explored using scatterplots
- Correlation heatmap created to show feature relationships

3.3 Statistical Analysis

One-Way ANOVA test was used to check whether fuel type affects selling price. The resulting p-value (less than 0.05) showed that fuel type significantly impacts car pricing.

3.4 Machine Learning Models

- K-Means clustering was used to group cars into:
  1. Cluster 0: Budget segment (old and highly used cars).
  2. Cluster 1: Mid-range segment.
  3. Cluster 2: Premium segment (new, less-used cars).

  Clustering was based on selling price, age, km driven, fuel type, and transmission.

4. Key Insights

- Older cars and highly used cars fall in the low-price cluster.
- Newer cars with low kilometres belong to the premium segment.
- Fuel type significantly affects pricing (Statistically proven using ANOVA).
- Most cars belong to the mid-range cluster indicating buyer demand.
- Mileage and age are the strongest determinants of resale value.

5.Conclusion

This project successfully analysed factors influencing used car prices using real-world data. The results show that price strongly depends on age, kilometres driven, and fuel type. K-Means clustering grouped cars into meaningful market categories. Statistical testing validated that fuel type significantly affects pricing. Future improvements could include adding engine specs, brand extraction, and applying advanced models for deeper insights